DISCOVERY & ASSESSMENT PLAN

Project: Northstar Outfitters — E-Commerce Platform Modernization

Reference: NSO-ECOM-2026-01 / Discovery Phase

Prepared by: Technical Consultancy Partner

Date: September 2026

1. Objectives

The primary objective of this Discovery & Assessment Phase is to evaluate Northstar Outfitters' current e-commerce platform, underlying architecture, and operational processes to establish an objective, data-driven modernization strategy.

Specifically, this phase aims to:

- Determine the technical health, security posture, and maintainability of the existing codebase without impacting live operations.
- Map all internal and external system integrations and evaluate their business criticality.
- Identify operational bottlenecks and manual workflows within operations and customer service teams.
- Provide executive management with a clear **Decision Matrix** comparing **Incremental Modernization** versus **Partial/Total Platform Replacement** in terms of feasibility, cost, time-to-market, and business risk.

2. Scope

In Scope:

- **Source Code & Database:** Static code analysis, dependency check, architectural pattern identification, and database structure review.
- **Integrations:** Mapping of all third-party systems (ERP, payment gateways, inventory management, logistics, CRMs, etc.).
- **Process & Workflows:** Operational interviews regarding order fulfillment, inventory sync, catalog management, and customer support procedures.
- **Non-Functional Requirements:** Security, reliability, scalability, and maintainability evaluation.

Out of Scope:

- Writing new production code or implementing functional changes to the existing store during this phase.
- Modifications, stress testing, or deployments directly on the live production environment.

3. Key Activities

[WEEKS 1 - 2] [WEEKS 2 - 3] [WEEK 4]

Technical Setup & Audit ---> Process & Integration ---> Final Evaluation &

Static Code Analysis Interviews & Mapping Decision Matrix

- **Environment Setup & Static Code Audit:** Setting up isolated local/QA environments and executing automated static code and vulnerability scans (e.g., SonarQube, Checkmarx).
- **System Integration Mapping:** Tracing data flows between the e-commerce core and external services to document undocumented connections.
- **Stakeholder Interviews:** Conducting structured interviews with internal IT, operations, and customer service staff.
- **Feasibility & Financial Synthesis:** Translating technical findings into business risks, calculating total effort estimates, and drafting the final recommendations.

4. Stakeholders & Interviews

To gain a holistic view of the platform, we propose short, structured sessions (45–60 mins each) with the following key personnel:

Role / Team	Focus / Objective
Internal IT Personnel (2)	Deep dive into system architecture, legacy code constraints, database logic, known bugs, and deployment procedures.
Operations & Fulfillment Team	Reviewing order processing, stock updates, return handling, and identifying manual workarounds.

Role / Team	Focus / Objective
Customer Service Team	Understanding common customer friction points during checkout, payment failures, or account issues.

5. Information and Access Required

To execute the assessment without delay, Northstar Outfitters will need to provide:

- Access to source code repositories (Git/Azure DevOps/SVN).
- Read-only access to a development/staging database copy (containing anonymized data if necessary).
- Existing documentation (if any), API specs, or third-party service credentials for QA/Dev environments.
- Contact points for key personnel to schedule interview sessions.

6. Assessment Criteria

Technical findings will be categorized into three business-oriented dimensions rather than purely abstract code metrics:

1. **Maintainability (Development Friction):** Measures how easy or difficult it is to introduce new features. High technical debt directly correlates with higher future development costs and slower time-to-market.
2. **Security & Reliability (Financial & Reputational Risk):** Identifies vulnerabilities, payment gateway stability issues, and single points of failure that could lead to downtime, data breaches, or lost revenue.
3. **Scalability (Growth Preparedness):** Evaluates if the current architecture can support anticipated traffic surges without performance degradation or system crashes.

7. Key Deliverables

At the conclusion of this phase, Northstar Outfitters will receive an **Executive Discovery Report** containing:

- **Technical Health & Security Audit:** Clear risk ratings (Low/Medium/High) covering code quality, security vulnerabilities, and platform stability.

- **Integration & Architecture Map:** A visual and descriptive inventory of all internal/external integrations and their business criticality.
- **Operational Friction Report:** Identified manual processes with recommendations for automation.
- **Executive Decision Matrix:** A structured comparison between **Option A (Incremental Modernization)** and **Option B (Platform Replacement)**, evaluating:
    - Estimated total cost and effort.
    - Timeline targeting the initial 6-month launch goal.
    - Operational and technical risk profiles.
    - Expected Return on Investment (ROI) and scalability over the next 3–5 years.

8. Timeline (4-Week Schedule)

Week	Focus	Main Deliverables / Milestones
Week 1	Access setup, local environment build, automated static code and security scanning.	Repository cloned; scanning tools configured; initial code metrics gathered.
Week 2	Technical IT interviews, integration tracing, and initial architectural mapping.	Integration inventory complete; IT technical debt report drafted.
Week 3	Operations & Customer Service interviews; manual workflow analysis.	Operational bottleneck report; business risk translation complete.
Week 4	Strategic synthesis, cost/effort modeling, and draft of Decision Matrix.	**Final Executive Discovery Report & Management Presentation.**

9. Assumptions and Constraints

- **Zero Operational Disruption:** All analysis, testing, and inspection will strictly occur in isolated local/QA environments. The live store will remain completely unaffected.
- **Information Availability:** Timely provision of access credentials and scheduling of interviews within Weeks 1 and 2 are essential to meet the 4-week delivery window.
- **Scope Refinement:** The outcomes of this discovery phase will serve as the baseline for defining the definitive scope, budget, and contract for the implementation phase.